

Production Code Observations

Separate from Theoretical Documents

Research Notes — May 2026

Purpose. This document records observations from reading the current production implementation (`system_mode.py`). It is kept separate from the theoretical derivations and the SIAM paper because the production code has been upgraded since the paper era, the BSDT/MFLS math implementation was fixed in a later update, and the exact correspondence between current code and paper-era code is uncertain. **Nothing here should be taken to modify or invalidate the theoretical documents.**

1 Architecture — Confirmed

The following are confirmed directly from code:

1. Three engines: `MolecularEngine` (LJ 6-12), `GravityModeEngine` (Gaussian/Coulomb), `HybridGravityEngine` ($w^* s_{\text{mol}} + (1 - w^*) s_{\text{grav}}$, weight by CV-AUROC).
2. **BSDT dual role.** During simulation: `BSDTChannels` instantiated as `bsdt_damper`, computing E_{BS} and ∇E_{BS} to drive the adaptive friction force (controller role). Post-simulation: `BSDTChannels` is one view in `FusedSystemScorer` (scorer role).
3. **UDL is post-simulation.** `FusedSystemScorer` runs on `X_final` (particle positions after convergence). The SIAM paper label “UDL Post-Simulation Operators” is exact. Input to the engine is raw `X`.
4. **Lyapunov decoupled from prediction.** Docstring: “Lyapunov stays inside the iteration loop only. Prediction signal is 100% topological/structural.”
5. **FusedSystemScorer.** Four views: `MorseTopologyAlarm` (4D, kNN), `BettiBarcodeSuite` (β_0, β_1, χ , Conley, 19D), `UDLPostSimScorer` (Phase+Topo+KernelRKHS+Rank, $\approx 27\text{D}$), `BSDTChannels` ($\delta_C, \delta_G, \delta_A, \delta_T$). Fused via Fisher Variance Ratio weights (data-determined).
6. **Fisher VR everywhere.** `MorseTopologyAlarm`, `BSDTChannels`, `FusedSystemScorer`, `ReducedTensorDescriptor` all use Fisher VR for weighting. No hardcoded weights.
7. **ReducedTensorDescriptor.** Implements Proposition 5.3 of the SIAM paper. $O(Nd + d^3)$ descriptor: grad norm, Hessian eigenvalues (via GMM), Mahalanobis, medoid distance, $\text{tr}(H)$, $\det(\Sigma)$, Morse index.

2 Differences from Paper-Era Description

2.1 Adaptive friction gain

SIAM paper describes (`MolecularEngine`): $\gamma^* = \alpha / \lambda_{\max}(\nabla^2 \Phi_{\text{pair}})$ via OGD with power iteration every 10 steps.

Current production code (both engines):

$$E_{\text{combined}} = E_{\text{BS}} + \beta_{\text{MFLS}} \cdot \text{MFLS}, \quad \gamma_{\text{prod}} = \frac{E_{\text{combined}}}{E_{\text{combined}} + \theta}$$

$\beta_{\text{MFLS}} = \theta_{\text{BS}}/\text{MFLS}_{\text{median}}$ (data-driven from normal-period reference). No OGD or power iteration present.

Status: Uncertain whether this reflects (a) an upgrade post-paper, (b) the BSDT/MFLS math fix, or (c) a separate architectural decision. Do not modify theoretical documents based on this observation alone.

2.2 BSDT/MFLS math fix

Author noted the BSDT/MFLS math implementation was corrected in a later update. Nature of the fix not yet recorded. Current code should be treated as the correct implementation. Whether pre-fix results in the SIAM paper match the current code is an open question.

3 Open Questions

1. What exactly was the BSDT/MFLS math fix?
2. Does E_{combined} represent the corrected gain formula, or is it a separate change from the paper-era γ_{κ}^* ?
3. Do the SIAM paper AUROC/lead-time results correspond to pre-fix or post-fix code?

4 What Is Not Affected by These Observations

The following theoretical results hold regardless of which gain formula is used in production:

Theorem C (Morse-index equivalence), Theorem 4.3 (phase transition), Lyapunov v2 convergence, geometric compression hierarchy (GC.1–GC.2a), probabilistic framework (Part VI), transition-type taxonomy, 34% MI loss result, and all empirical validation results.